

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

SCIENCE

0893/02

Paper 2

April 2025

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

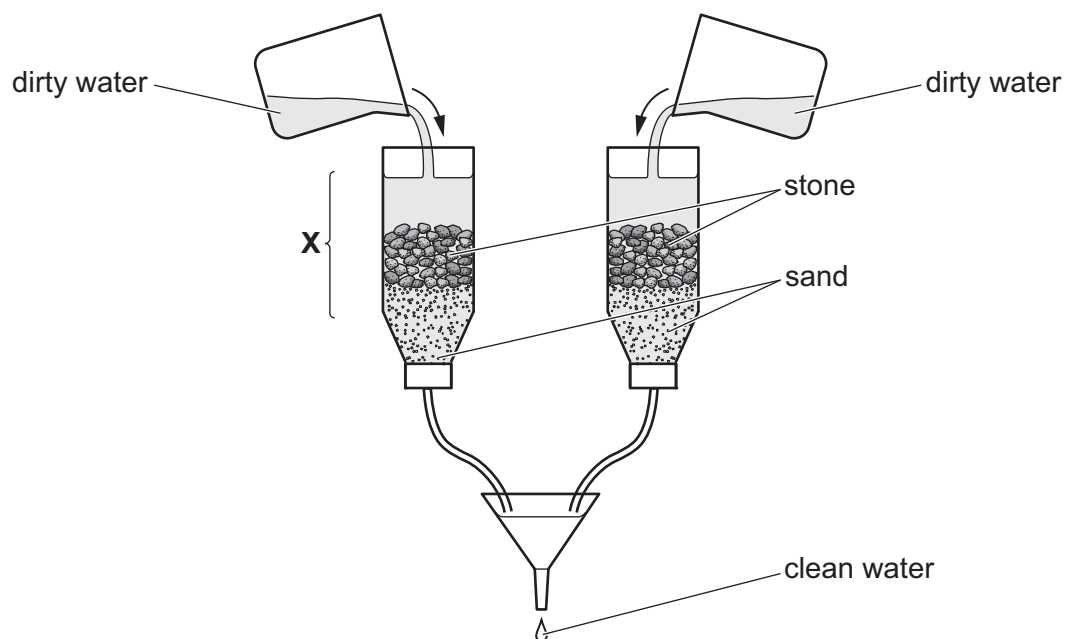
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

1 Look at this model of the human excretory (renal) system.



(a) (i) What organ does **X** represent?

X [1]

(ii) Name the liquid excreted by the renal system.

..... [1]

(b) Describe **one strength** and **one limitation** of this model.

strength

.....

limitation

.....

[2]

(c) Suggest **one** reason why scientific models change over time.

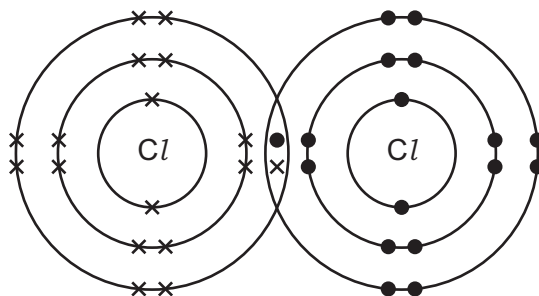
.....

..... [1]

- 2 Chlorine is an element in Group 7 of the Periodic Table.

Chlorine forms a molecule with a covalent bond.

Look at the model of a molecule of chlorine.



- (a) What do the crosses and dots represent in this model?

..... [1]

- (b) Explain why bromine forms a molecule with a similar structure to chlorine.

Use ideas about the Periodic Table.

.....
 [1]

- (c) Look at the symbol for a particle.



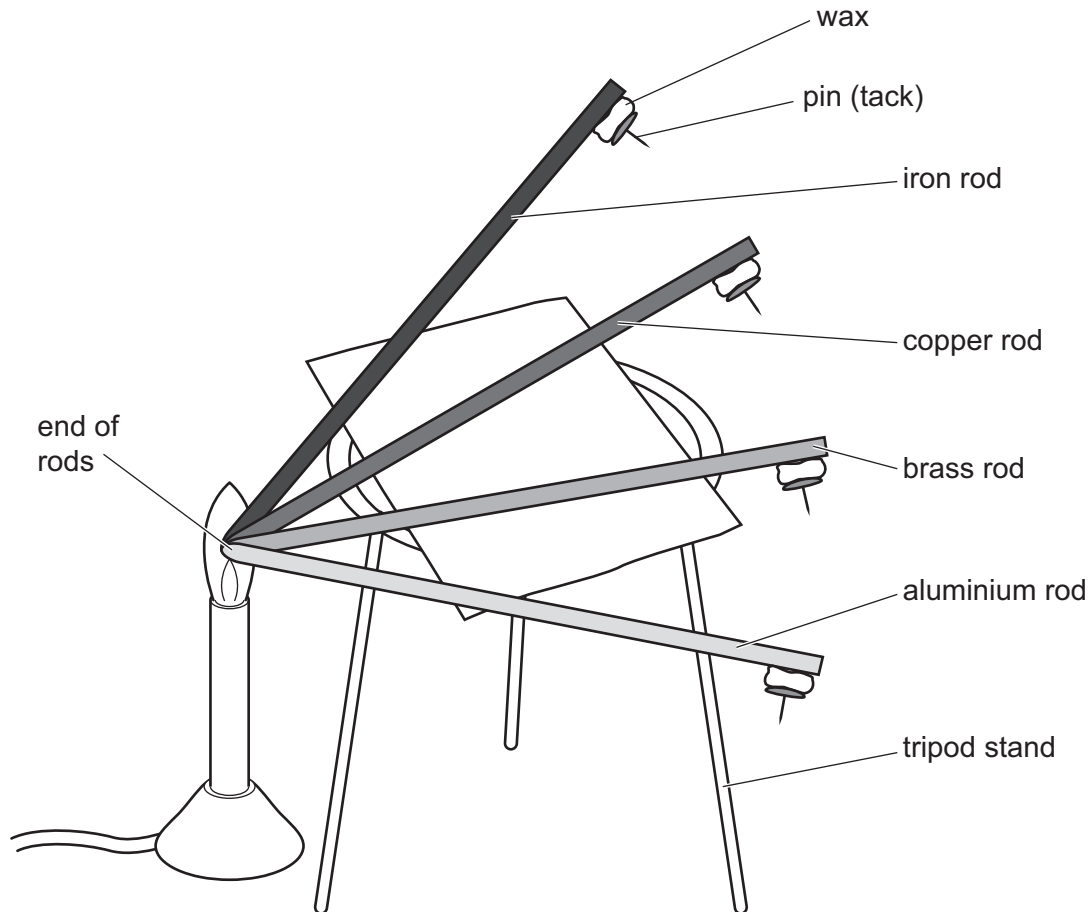
Explain how this particle is made from a chlorine atom.

.....
 [1]

3 Aiko investigates conduction.

Aiko:

- attaches a pin (tack) to four different metal rods with wax
- assembles the equipment shown in the diagram



- heats one end of the metal rods
- measures the time it takes for the wax to melt and the pin (tack) to fall from each metal
- records these times in a table.

Look at her results.

metal	time taken for pin (tack) to fall in seconds
iron	62
copper	19
brass	32
aluminium	48

(a) Which metal in the table is the best thermal conductor?

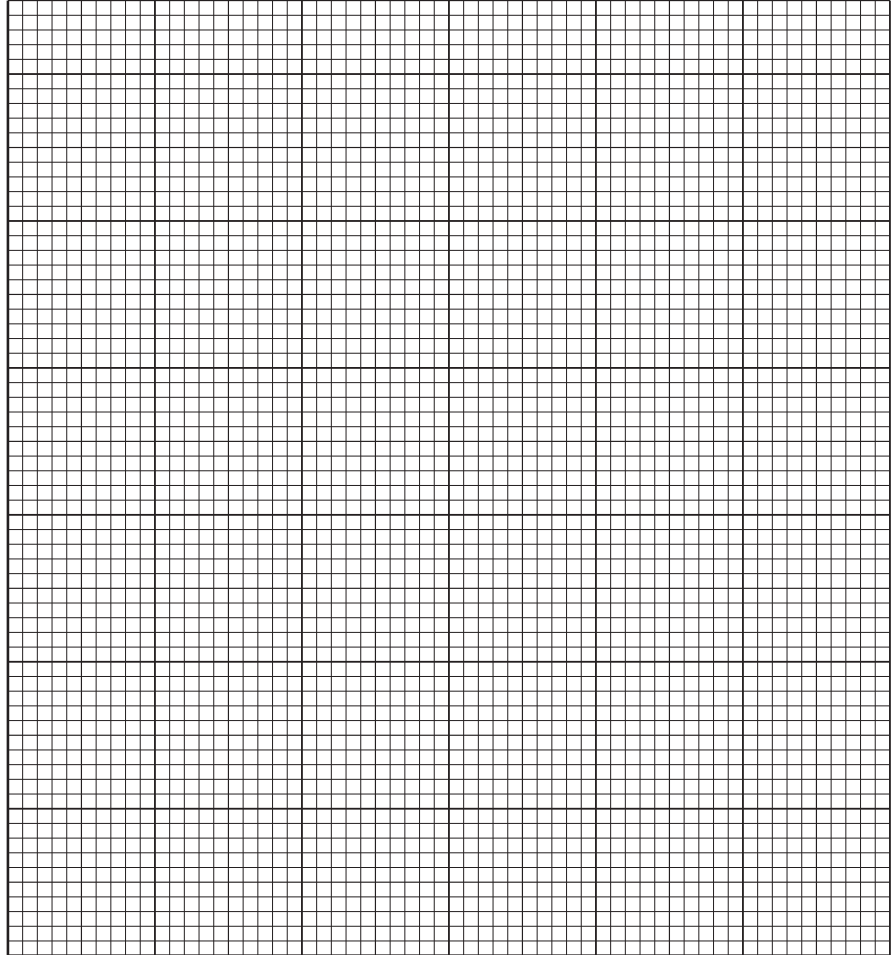
..... [1]

(b) Draw a bar chart of the time taken for each pin (tack) to fall in seconds.

Include:

- labels for the x -axis and the y -axis
- a suitable scale for the y -axis
- bars for the four metals.

.....



.....

[3]

4 This question is about the link between climate change and global temperature.

(a) Write down **two** impacts of climate change.

1

2

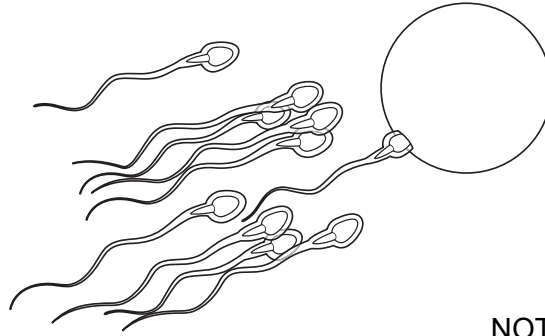
[2]

(b) The increase in global temperature is caused by increasing levels of carbon dioxide in the atmosphere.

Name **one** process in the carbon cycle that releases carbon dioxide into the atmosphere.

..... [1]

- 5 The diagram shows two types of sex cells involved in human reproduction.



NOT TO SCALE

- (a) What is the name for these types of sex cells?

..... [1]

- (b) The nucleus of each cell contains chromosomes.

Complete the sentence.

Chromosomes contain made of

..... [1]

- (c) The inheritance of sex in humans is determined by a pair of sex chromosomes.

There are two different sex chromosomes, X and Y.

The diagram shows the inheritance of X and Y chromosomes.

		parent 1	
		X	X
parent 2	X	XX	XX
	Y	XY	XY

Parent 1 and parent 2 have a baby.

What is the probability that the baby is a girl?

.....

Explain your answer.

.....

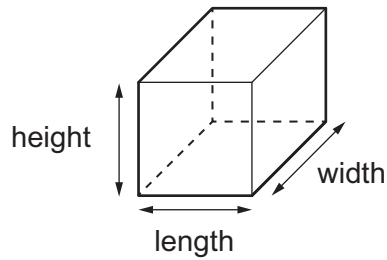
.....

.....

[2]

- 6 Rajiv investigates the density of three different blocks of metal.

Look at the diagram of one of the blocks of metal.



- (a) Rajiv records data for the height, length and width of the three blocks of metal.

metal A
height is 3 cm, length = 3 cm, width = 2 cm
metal B
length = 1.5 cm, width = 3 cm, height is 2.5 cm
metal C
width = 4 cm, length = 3 cm, height is 4 cm

Write this data in a suitable results table.

Include the headings in your table.

- (b) The volume of another block of metal is 11 cm^3 .

The mass of this block is 88 g.

Calculate the density of this block.

Include the unit of density in your answer.

density = unit [3]

- (c) Rajiv measures the volume of three different objects.

Each object has the **same** mass.

Look at the table.

object	volume in cm^3
D	16.3
E	21.4
F	5.2

Predict the order of **density** of the objects from **least** dense to **most** dense.

least dense



most dense

.....

.....

.....

[1]

- 7 A lamp transfers 200 J of electrical energy into 120 J of useful light energy.

The lamp dissipates some energy into the surroundings.

- (a) Calculate the amount of energy dissipated into the surroundings.

amount of energy dissipated = J [1]

- (b) What type of energy is dissipated?

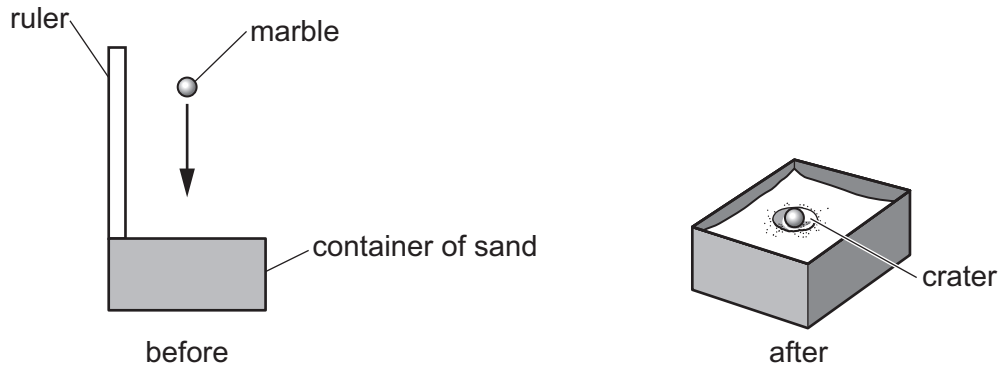
..... [1]

- (c) Calculate the percentage of electrical energy transferred into useful light energy.

percentage = % [2]

- 8 Mia uses a model to investigate asteroid collisions with the Earth.

Look at a diagram of her assembled equipment.



In her first experiment Mia:

- drops a marble from a height of 1 m into the sand
- measures the diameter of the crater made in the sand
- records the result in a table.

Mia repeats the experiment four more times.

Each time she uses a marble with a different mass.

Look at her results.

mass of marble in g	diameter of crater in mm
4	30
6	42
8	59
10	72
12	98

- (a) Before the experiment Mia makes a prediction.

The results support her prediction.

Suggest what Mia predicts.

.....
 [1]

(b) Mia drops the marble from the same height.

Write down **one other** variable Mia keeps the same.

..... [1]

(c) Suggest **two** possible effects of a large asteroid colliding with the Earth.

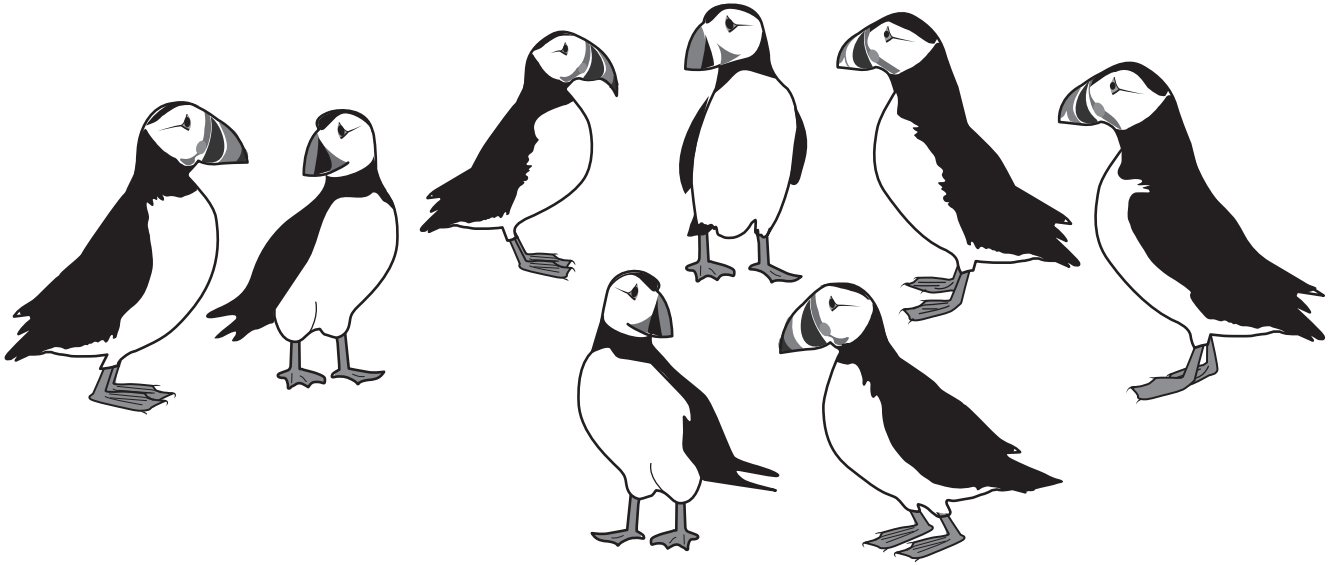
1

2

[2]

9 The diagram shows the variation in a group of puffins.

These variations cause different characteristics.



Variation is caused by:

- genes
- the environment
- a combination of genes and the environment.

(a) Write down **one** characteristic shown in the diagram that is determined **only** by genes.

..... [1]

(b) Puffins feed on a fish called hake.

The hake are moving into deeper waters because the sea is becoming warmer.

Puffins find it difficult to catch hake in the deeper water.

The sea becoming warmer affects the population of puffin and the population of hake.

(i) Describe the effect on the hake population.

..... [1]

(ii) Explain what happens to the population of puffin if the sea continues to warm.

.....
 [2]

10 Group 1 elements in the Periodic Table are metals.

(a) Look at the table showing the melting points of some Group 1 elements.

element	melting point in °C
lithium	181
sodium	
potassium	64
rubidium	39

Predict the melting point of sodium.

..... °C [1]

(b) A teacher demonstrates the reaction between sodium and water.

This reaction makes sodium hydroxide and hydrogen.

Mia observes:

- sodium moving very quickly over the surface of the water
- a yellow flame.

(i) Suggest **one** way the teacher demonstrates this reaction safely.

.....
..... [1]

(ii) Mia also observes the reaction of potassium with water.

Write down **one** observation that is **different** for this reaction compared to the reaction of sodium and water.

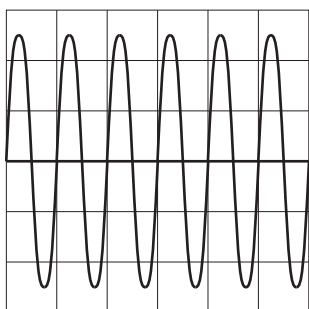
.....
..... [1]

(iii) Write down the **word** equation for the reaction of potassium with water.

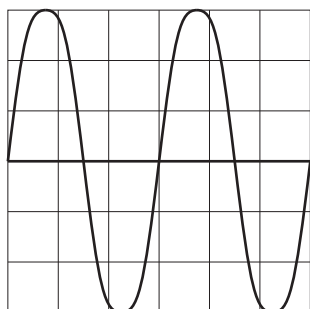
..... [1]

11 Lily investigates the waveforms of sounds produced by musical instruments.

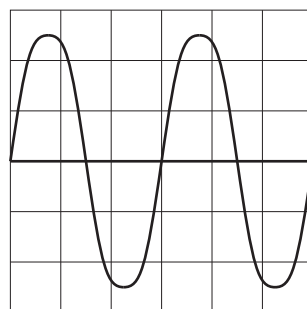
(a) Look at the waveforms.



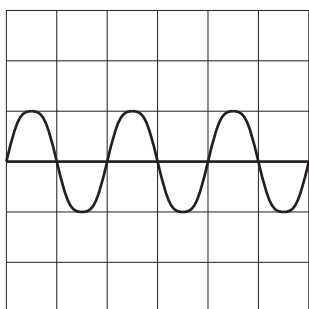
A



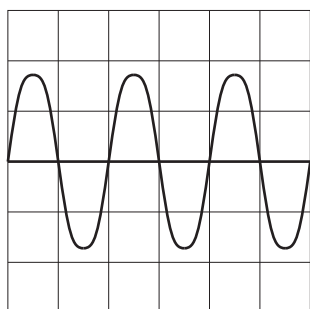
B



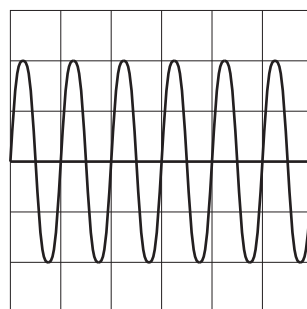
C



D



E



F

(i) Which **two** waveforms have the same **amplitude**?

..... and

[1]

(ii) Which **two** waveforms have the highest **frequency**?

..... and

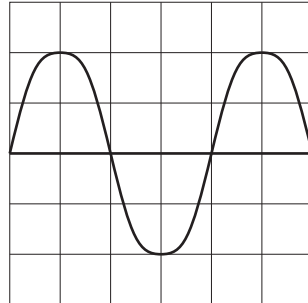
[1]

(b) Sound waves interact with each other by cancelling out or reinforcing.

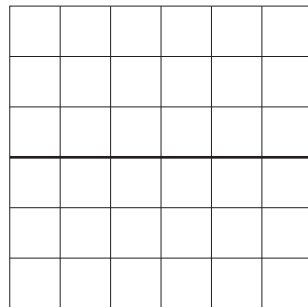
(i) Wave **G** has the waveform **G**.

Wave **H** completely cancels wave **G**.

Draw waveform **H** on the blank grid.



G



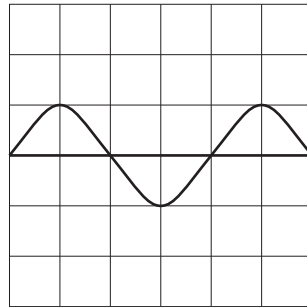
H

[2]

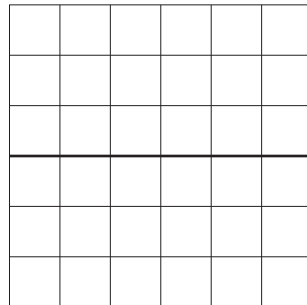
(ii) Wave **J** has the waveform **J**.

Wave **K** completely reinforces wave **J**.

Draw waveform **K** on the blank grid.



J



K

[1]

12 This question is about the life of stars.

Complete the sentences to explain how stars are formed.

A nebula is a giant cloud of and

A nebula is referred to as a because it is where stars are formed.

[3]

BLANK PAGE

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

The Periodic Table of Elements

Group																							
1	2						3	4	5	6	7	8											
1 H hydrogen 1												2 He helium 4											
3 Li lithium 7 4 Be beryllium 9												5 B boron 11 6 C carbon 12 7 N nitrogen 14 8 O oxygen 16 9 F fluorine 19 10 Ne neon 20											
11 Na sodium 23 12 Mg magnesium 24												13 Al aluminium 27 14 Si silicon 28 15 P phosphorus 31 16 S sulfur 32 17 Cl chlorine 35.5 18 Ar argon 40											
19 K potassium 39 20 Ca calcium 40												21 Sc scandium 45 22 Ti titanium 48 23 V vanadium 51 24 Cr chromium 52 25 Mn manganese 55 26 Fe iron 56 27 Co cobalt 59 28 Ni nickel 59 29 Cu copper 64 30 Zn zinc 65 31 Ga gallium 70 32 Ge germanium 73 33 As arsenic 75 34 Se selenium 79 35 Br bromine 80 36 Kr krypton 84											
37 Rb rubidium 85 38 Sr strontium 88												39 Y yttrium 89 40 Zr zirconium 91 41 Nb niobium 93 42 Mo molybdenum 96 43 Tc technetium — 44 Ru ruthenium 101 45 Rh rhodium 103 46 Pd palladium 106 47 Ag silver 108 48 Cd cadmium 112 49 In indium 115 50 Sn tin 119 51 Sb antimony 122 52 Te tellurium 128 53 I iodine 127 54 Xe xenon 131											
55 Cs caesium 133 56 Ba barium 137												57–71 lanthanoids 72 Hf hafnium 178 73 Ta tantalum 181 74 W tungsten 184 75 Re rhenium 186 76 Os osmium 190 77 Ir iridium 192 78 Pt platinum 195 79 Au gold 197 80 Hg mercury 201 81 Tl thallium 204 82 Pb lead 207 83 Bi bismuth 209 84 Po polonium — 85 At astatine — 86 Rn radon —											
87 Fr francium — 88 Ra radium —												89–103 actinoids 104 Rf rutherfordium — 105 Db dubnium — 106 Sg seaborgium — 107 Bh bohrium — 108 Hs hassium — 109 Mt meitnerium — 110 Ds darmstadtium — 111 Rg roentgenium — 112 Cn copernicium — 113 Nh nihonium — 114 Fl flerovium — 115 Mc moscovium — 116 Lv livermorium — 117 Ts tennessine — 118 Og oganeson —											
lanthanoids																							
actinoids																							